



Interpretable Multi-Source Machine Learning for Urban Classification in Data-Scarce Regions



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PANEL 2

BACKGROUND & MOTIVATION

Rapid Urbanization
Rapid urbanization is transforming cities across the Global South, placing increasing pressure on infrastructure, land resources, and sustainable development.

Challenges in Urban Classification
Although remote sensing and machine learning have greatly improved urban mapping, accurate classification remains challenging in data-scarce regions due to limited labeled data, heterogeneous geospatial information, and the limited interpretability of many AI models.

Opportunity with Explainable AI
Explainable Artificial Intelligence (XAI) enables transparent machine learning by identifying the factors driving model predictions, thereby improving trust and supporting evidence-based urban planning.

Interpretable Multi-Source Machine Learning Framework
This study proposes an interpretable multi-source machine learning framework for urban classification using Lagos, Nigeria as a case study.

PANEL 3

RESEARCH GAP & OBJECTIVES

Research Gap

- Limited Data Integration**
Many urban classification studies rely mainly on spectral imagery while underutilizing complementary socio-economic and temporal geospatial datasets.
- Limited Interpretability**
Although machine learning models achieve high classification accuracy, they often provide limited insight into the factors influencing their predictions.
- Limited Evidence from Africa**
Relatively few interpretable urban classification studies have been conducted in rapidly growing, data-scarce African cities such as Lagos.
- Fragmented Frameworks**
Existing approaches rarely integrate multi-source data, machine learning, and explainable AI within a single transparent framework.

Proposed Framework & Research Objectives

Proposed Framework

Research Objectives

- Integrate heterogeneous geospatial datasets for urban classification.
- Develop and compare Random Forest, XGBoost, and Neural Network models.
- Evaluate predictive performance using Accuracy and ROC-AUC metrics.
- Interpret model predictions using SHAP explainable AI.
- Identify key drivers of urbanization to support transparent urban planning.

This study presents an interpretable multi-source machine learning framework that integrates heterogeneous geospatial data with explainable AI to deliver accurate, transparent, and scalable urban classification for data-scarce regions.

PANEL 4

MATERIALS & METHODS

1. STUDY CONTEXT
This study was conducted in Lagos, Nigeria, one of Africa's fastest-growing megacities, characterized by rapid urban expansion and heterogeneous land cover dynamics. The goal is to develop an interpretable machine learning framework for urban classification in data-scarce environments.

2. MULTI-SOURCE GEOSPATIAL DATASET

Source	Description	Variables / Products	Spatial Resolution	Temporal Coverage
LandSat 5/7/9	Optical satellite imagery	Blue, Green, Red, NIR, SWIR (5 Bands)	30 m	2000-2024
GHSL	Global Human Settlement Layer	Built-up / Urban Labels	30 m	2000-2024
VIIRS DNB	Night-time lights	Night-time Light Intensity	500 m	2012-2024
WorldPop	Population density	People per km ²	100 m	2012-2024
SRTM DEM	Topographic data	Elevation (m)	30 m	Static
Time	Temporal dimension	Year	-	2000-2024

3. DATA PROCESSING WORKFLOW

4. MACHINE LEARNING PIPELINE

A harmonized multi-source geospatial dataset comprising spectral, socio-economic, topographic, and temporal variables was developed to train interpretable machine learning models for urban classification.

PANEL 5

RESULTS: MODEL PERFORMANCE

1. ROC CURVE COMPARISON FOR URBAN CLASSIFICATION

Figure 1. ROC curves comparing the three models for urban classification. XGBoost achieved the highest AUC (0.943), followed by Random Forest (0.942) and Neural Network (0.941).

2. RANDOM FOREST FEATURE IMPORTANCE

Figure 2. Random Forest feature importance showing that the Green spectral band, night-time lights, and population density are the most influential predictors of urban classification.

3. MODEL PERFORMANCE SUMMARY

Model	Accuracy	AUC (ROC)
Random Forest	0.87	0.942
XGBoost	0.87	0.943
Neural Network	0.87	0.940

4. KEY TAKEAWAYS

- Consistent High Performance:** All three models achieved an accuracy of 0.87 and AUC > 0.94, demonstrating strong discriminative performance for urban classification.
- Best Discriminative Ability:** XGBoost achieved the highest AUC (0.943), indicating excellent ability to distinguish urban from non-urban areas.
- Feature Representation Matters:** Ensemble models (Random Forest and XGBoost) performed comparably to the neural network, highlighting that feature representation is more critical than model complexity.

The results demonstrate that interpretable machine learning models, driven by multi-source geospatial features—particularly night-time lights and population density—can effectively support accurate urban classification.

PANEL 6

EXPLAINABLE AI (SHAP): UNDERSTANDING MODEL DECISIONS

1. SHAP SUMMARY (BEEHIVE) PLOT

Figure 3. SHAP summary (beehive) plot showing the distribution of SHAP values for all features. Each dot represents a sample colored by the feature value (blue = low, red = high).

2. SHAP FEATURE IMPORTANCE (MEAN SHAP)

Figure 4. Mean absolute SHAP value for each feature. Night-time lights and population density are the most important predictors of urban classification.

3. SHAP DEPENDENCE PLOT: NIGHTLIGHTS

Figure 5. SHAP dependence plot for Nightlights. As night-time lights increase, SHAP values increase, indicating a higher contribution to urban classification. Color shows the interaction with SWIR.

4. SUMMARY OF SHAP ANALYSIS

Feature	Direction of Impact on Urban Probability	Interpretation
Night-time Lights	↑	Higher values strongly increase the probability of an area being classified as urban.
Population Density	↑	Higher population density strongly increases urban probability.
NDVI	↓	Built-up surfaces contribute positively to urban prediction.
Green (Band)	↓	Higher vegetation decreases the probability of an area being urban.
NDVI	↓	Higher NDVI (greenness) reduces urban likelihood.

5. KEY INSIGHTS FROM SHAP

- Night-time lights is the most influential predictor, followed by population density and built-up index (NDBI).
- Higher values of Night-time lights and population density consistently increase the likelihood of urban areas.
- Vegetation-related features (Green, NDVI) have negative SHAP values, indicating lower probability of urban classification.
- SHAP explanations enhance interpretability and support data-driven, transparent urban planning and decision-making.

SHAP analysis reveals that Night-time lights and population density are the dominant drivers of urban prediction, while vegetation metrics reduce urban probability—providing transparent insights into model behavior.

PANEL 6B

URBAN PROBABILITY MAP OF LAGOS (2020)

Figure 6 illustrates the spatial distribution of predicted urban probability across Lagos State for 2020. The map highlights the heterogeneous nature of urban development, with higher probabilities concentrated within the metropolitan core and lower probabilities toward peri-urban and less-developed regions.

Urban Probability Map of Lagos (2020)

KEY INFORMATION

- Study Area: Lagos State, Nigeria
- Prediction Year: 2020
- Prediction Model: XGBoost (Highest ROC-AUC)
- Spatial Resolution: 100 meters
- Validation: ROC-AUC = 0.943
- Visualization: Relative Probability (2-98 percentile stretch)

HIGHLIGHTS

- Urban probability is highest within the Lagos metropolitan core (mainland, Lagos Island and Ibeju axis).
- Spatial gradients reveal heterogeneous urban expansion patterns.
- Peri-urban areas exhibit intermediate urban probability, while rural areas show the lowest probabilities.
- The probability surface supports spatially explicit urban planning and monitoring.

NOTES

- Map derived from predicted urban probability using the test dataset for the year 2020.
- Urban classification threshold probability > 0.50 (used for evaluation only, not displayed on map).

PANEL 7

CONCLUSIONS, FUTURE WORK & AVAILABILITY



DLI 2026
DEEP LEARNING INDABA



CONCLUSIONS

- ✓ We developed an interpretable multi-source machine learning framework for urban classification in data-scarce regions.
- ✓ Ensemble models achieved high predictive performance with Accuracy = 0.87 and ROC-AUC > 0.94 across all models.
- ✓ SHAP explainability identified night-time lights, population density and the Green spectral band as the dominant drivers of urban classification.
- ✓ The framework supports transparent, scalable and data-driven urban monitoring in rapidly developing cities.



KEY FINDINGS

- ✓ Green spectral band is the most important predictor in the Random Forest model.
- ✓ Night-time lights show the strongest positive contribution in SHAP analysis.
- ✓ Population density strongly influences urban probability and complements spectral variables.
- ✓ Feature integration is more critical than model complexity, with ensemble methods performing comparably to deep learning models.



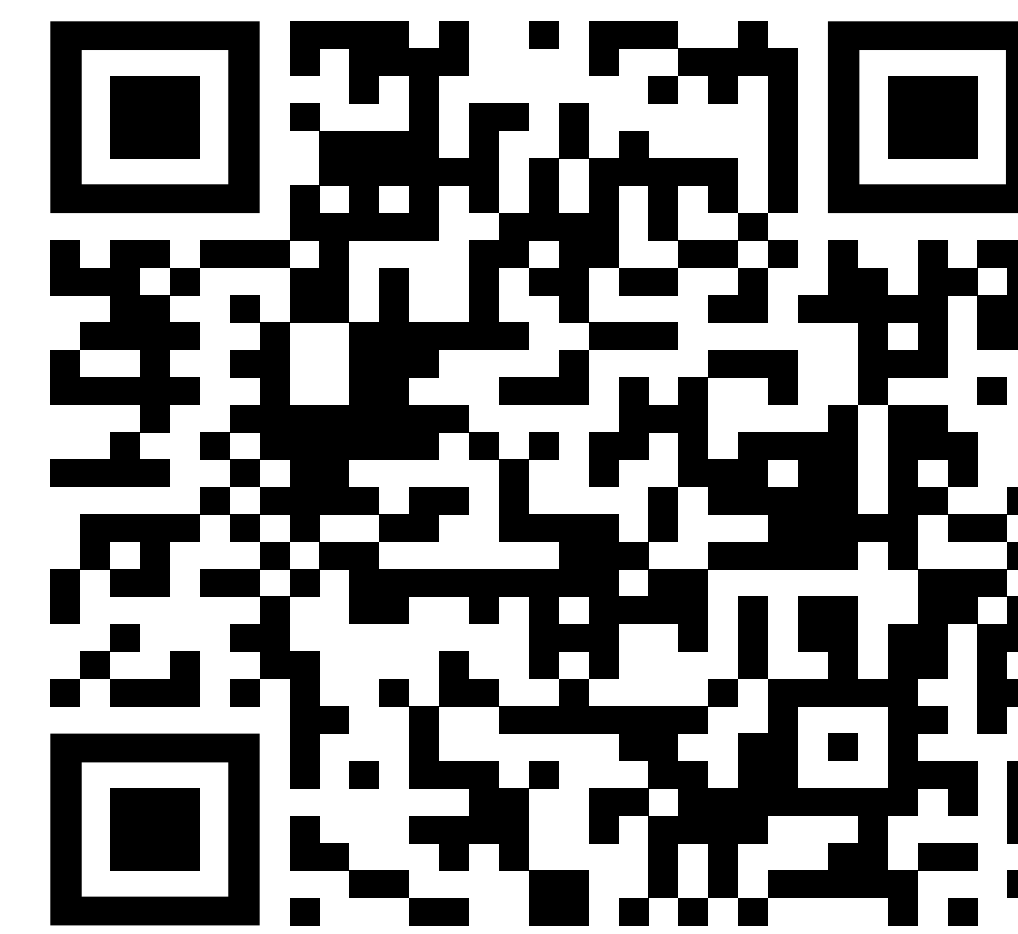
FUTURE WORK

- ✓ Integrate spatial context using Convolutional Neural Networks (CNNs).
- ✓ Investigate spatially explicit deep learning for improved urban classification.
- ✓ Extend the framework to other African cities for comparative urban studies.
- ✓ Support operational urban monitoring and decision support systems.



PAPER • CODE • DATA

Scan to access the paper, code, figures and reproducible workflow.



GitHub Repository
github.com/ugepet/interpretable-urban-classification



Direct Link
<https://github.com/ugepet/interpretable-urban-classification>



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RESEARCH IMPACT

Interpretable AI combined with heterogeneous geospatial data provides an **accurate**, **transparent** and **scalable** framework for urban monitoring and decision support in rapidly developing cities.

